

Fatema Tuj Johora Faria

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Research Interests

- **AI Reasoning & Autonomy:** LLM Reasoning, AI Agents, Multimodal AI
- **Foundations of Dependable AI:** AI Safety, AI Security, Trustworthy AI
- **Domain-Specific AI:** Vision-Language Models for Medical Imaging, AI for Databases (NL-to-SQL)

Education

Ahsanullah University of Science and Technology (AUST)

July 2019 – December 2023

B.Sc. in Computer Science and Engineering (CGPA: 3.302 / 4.00)

Dhaka, Bangladesh

- **Undergraduate Thesis:** Investigated generative adversarial networks (CycleGAN and Pix2Pix) for potato disease image generation to address limited training data, and evaluated the generated images for disease classification using MobileNetV2, DenseNet169, and InceptionResNetV2, and for instance-level segmentation with Mask R-CNN.
- **Thesis Supervisor:** Prof. Dr. Mohammad Shafiul Alam
- **Thesis Publications:** [IEEE CCWC 2023](#) & [IEEE i-COSTE 2025](#)

Publications

Workshops

- Fatema Tuj Johora Faria, M. B. Moin, J. A. Mahmud, M. F. Mridha, and M. A. Hossain. “CLAIR-Fin: An Adversarial Multi-Agent Framework for Claim-Level Verification and Adaptive Debate in Cross-Modal Financial QA,” *arXiv preprint arXiv:2608.13706*, 2026. [\[Accepted at FinNLP-2026 @ EMNLP-2026\]](#) 🔗
- Fatema Tuj Johora Faria, M. B. Moin, J. A. Mahmud, M. F. Mridha, and M. A. Hossain. “LabourCrew: A Multi-Agent RAG Framework for Trustworthy Adversarial Deliberation and Statutory Reasoning over Labour Law,” 2026. [\[Submitted to Natural Legal Language Processing 2026 @ EMNLP-2026\]](#)
- Fatema Tuj Johora Faria, M. B. Moin, M. M. Rahman, K. M. Hasib, J. A. Mahmud, and M. F. Mridha. “Guiding Language Models to Be More Empathetic: Culturally Sensitive Mental Health Advice Generation Through Human-LLM Collaboration,” *arXiv preprint arXiv:2607.23538*, 2026. 🔗

Conferences

- Fatema Tuj Johora Faria, M. B. Moin, M. F. Mridha, and J. A. Mahmud. “TeachMateGPT: A Multi-Agent Knowledge-Grounded Framework for Pedagogical Assessment Generation from Science Curriculum Materials,” *arXiv preprint arXiv:2608.13708*, 2026. [\[Submitted to EACL 2027\]](#) 🔗
- E. Hossain, M. M. H. Nipu, Fatema Tuj Johora Faria, T. N. Ornee, and M. Sheikh. “ChannelGuard: Safe Models Do Not Compose into Safe Multi-Agent Systems,” *arXiv preprint arXiv:2607.19430*, 2026. [\[Submitted to SIGKDD 2027 Research Track\]](#) 🔗

Journals

- Fatema Tuj Johora Faria, M. B. Moin, Z. Hasan, M. A. A. Khandaker, N. Islam, K. M. Hasib, and M. F. Mridha. “Multi-BanFakeDetect: Integrating Advanced Fusion Techniques for Multimodal Detection of Bangla Fake News in Under-Resourced Contexts,” *International Journal of Information Management Data Insights*, 5(2), 100347, 2025. 🔗
- Fatema Tuj Johora Faria, M. B. Moin, B. K. Rafa, S. Saha, M. M. Rahman, K. M. Hasib, and M. F. Mridha. “Bangla-CalamityMMD: A Comprehensive Benchmark Dataset for Multimodal Disaster Identification in the Low-Resource Bangla Language,” *International Journal of Disaster Risk Reduction*, 130, 105800, 2025. 🔗
- Fatema Tuj Johora Faria, Mukaffi Bin Moin, Khan Md Hasib, Md Rafiqul Islam, and Sami Azam. “HEAL: A Self-Correcting Multi-Agent Framework with Mutual Consistency Validation for Clinical Text-to-SQL,” 2026. [\[Manuscript in Preparation for Expert Systems with Applications\]](#)
- Fatema Tuj Johora Faria, M. B. Moin, A. A. Wase, M. Ahmmed, M. R. Sani, and T. Muhammad. “Vashantor: A Large-Scale Multilingual Benchmark Dataset for Automated Translation of Bangla Regional Dialects to Bangla Language,” *arXiv preprint arXiv:2311.11142*, 2025. [\[Under Review in Language Resources and Evaluation\]](#) 🔗

Professional Experience

Astha IT

AI Engineer II

November 2025 – Present

Dhaka, Bangladesh

Product: TalentX | Python, LangGraph, LangChain, OpenAI, PostgreSQL (pgvector), WebSocket, FastAPI, React, GCP Services

- Built a “Job Builder Agent” that combines document parsing, field-level extraction, and human-in-the-loop refinement to transform user requirements or PDF/DOCX inputs into structured, editable job specifications.
- Designed a “Profile Screening Agent” that performs resume information extraction, career-page parsing, semantic candidate–job matching, and ranking to identify qualified candidates against job-specific requirements.
- Developed a “Job Fit Scoring Agent” that dynamically generates evaluation criteria and weights from job requirements, applies weighted scoring and requirement-level candidate assessment, and sends HRs detailed evaluation reports with interpretable scores and evidence-based explanations via email.
- Architected a “Talent Matching Agent” that converts HR queries into structured retrieval constraints and combines semantic vector search, metadata filtering, query optimization, and relevance reranking to retrieve evidence-backed candidate recommendations, with dynamic removal of outdated profiles and ingestion and updating of new profiles.
- Optimized large-scale CV processing through batch execution, parallel processing, and redundant LLM-call reduction, enabling the evaluation of 800–1,000 candidate profiles and **cutting manual effort from 5 days to under 12 hours**.

Dexian (Bangladesh) Limited

Senior Application Developer

July 2025 – October 2025

Dhaka, Bangladesh

Project: ShareFlow Agent | Python, LlamaIndex, Azure OpenAI, Microsoft Azure Services, FastAPI, RAGAS, TypeScript, React

- Architected a multi-agent RAG system integrating SharePoint and web sources via dynamic tool invocation, leveraging ReAct reasoning and source-aware context construction for knowledge-grounded responses.
- Developed a “Retrieval Agent” with ReAct planning, query reformulation, retrieval filtering, and custom guardrails to constrain responses to internal documents and reduce unsupported generation.
- Orchestrated an “OCR Agent” to extract content from diverse unstructured sources (scanned PDFs, images, and DOCX files) with rate limiting to support fault-tolerant performance.
- Integrated a “Web Browser Agent” with controlled web retrieval and instruction-aware source selection to augment knowledge-grounded responses with current external information when enterprise sources were insufficient.
- Implemented dynamic knowledge-base management with incremental SharePoint synchronization, file-level tracking, and memory updates, allowing users to add or remove knowledge sources without rebuilding the complete index.
- Designed session-aware conversational workflows with persistent history, context management, and session reset mechanisms, and integrated the system with Microsoft Bot Framework for deployment within Microsoft Teams.
- Directed system architecture and mentored junior developers through design reviews, debugging, code reviews, and technical knowledge-sharing, achieving an **average 60–65% annual operational cost reduction** through agent and inference optimization for 80 Sales Managers with 50+ interactions per day.

Dexian (Bangladesh) Limited

Application Developer

May 2024 – June 2025

Dhaka, Bangladesh

Project: Org Info | Python, LangGraph, LangChain, Azure OpenAI, Microsoft Azure Services, React, FastAPI, OpenCV, WebSocket

- Implemented an “OrgChart Extraction Agent” with Tree-of-Thought (ToT) reasoning, visual parsing, hierarchical inference, and structured entity mapping to reconstruct organizational hierarchies from organograms and map extracted roles to corresponding Bullhorn records in a relational database.
- Developed an “OrgQuery Agent” that converts natural language queries into optimized SQL (text-to-SQL) using Chain-of-Thought (CoT) and self-consistency prompting, retrieves organizational hierarchy from the relational database, and presents results through the OrgChart interface for hierarchical visualization.
- Engineered “OrgInfo Assistant” to allow users to interact with specific organizational hierarchies by converting queries to SQL, executing them on a temporary organization-specific database to reduce load on the primary database, and converting the retrieved results to natural language.
- Created an “OrgActivity Summarization Agent” to generate 7-day summaries of organizational activities by extracting relevant notes on placements, submissions, and communication logs from the Bullhorn database.
- Automated Bullhorn synchronization through scheduled data ingestion and incremental database updates, **cutting organizational hierarchy search time by 75%** by eliminating repeated direct queries against the source system.

Project: RFPMatcher | Python, LlamaIndex, Azure OpenAI, Microsoft Azure Services, CouchDB, React, FastAPI, WebSocket

- Architected a document-centric RAG pipeline with Chain-of-Thought reasoning and 12 specialized analytical modules to extract key information (e.g., project scope and deliverables) from Request for Proposal (RFP) documents.
- Orchestrated a master RFP database by processing historical responses with metadata (generated answers, summaries, win/loss labels, and question categories) and vector embeddings to support semantic search.
- Designed a “Past Experience Matcher” system that analyzes new RFP documents by extracting key client requirements, generating binary (Yes/No) and descriptive questions for each section, and querying a master database to identify relevant past experiences through semantic similarity matching.
- Developed a Table of Contents (TOC) generation workflow that analyzes extracted RFP key information against pre-defined section libraries to intelligently select and prioritize relevant sections based on project requirements.
- Implemented a “Proposal Writing Assistant” that generates initial content for each TOC section using extracted RFP information and relevant past experience, enabling iterative refinement through targeted chat interactions.
- Accelerated decision-making through predictive insights, with **manual review of 100+ pages reduced by 3–5 days** and Proposal Managers able to focus on strategic bid development.

Academic Reviewing

Academic Peer Reviewer

2025 – Present

Journals & Workshops

- **Journals:** Scientific Reports; Language Resources and Evaluation; Discover Mental Health; Cluster Computing
- **Workshop:** FinNLP-2026 @ EMNLP-2026

Ongoing Research Projects

- Investigating Positional Bias in Vision-Language Models through Medical Image Segmentation and Visual Localization
- Script, Mixing, or Switching? Disentangling the Causes of Linguistic-Form Fairness Gaps Across Multi-Turn LLM Tutoring Conversations

Technical Skills

- **Programming Languages:** Python, Java, C++
- **Web Development:** JavaScript, TypeScript, React, Tailwind CSS, FastAPI
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Deep Learning Frameworks:** TensorFlow, Keras, PyTorch
- **LLM Application Frameworks:** LangChain, LangGraph, LlamaIndex
- **Cloud Services:** Azure (Azure OpenAI, Azure SQL Database, Azure Blob Storage, Azure App Service); AWS (Amazon ECR, Amazon EC2, Amazon S3); GCP (Google Cloud Storage, Google App Engine, Google Compute Engine)
- **Others:** AlloyDB for PostgreSQL (pgvector), Milvus, Docker, Microsoft Bot Framework, OpenCV, WebSocket

Awards & Achievements

- Received a **50% registration fee scholarship** for research excellence and presentation of “*Generative Adversarial Networks for Potato Disease Classification and Instance Segmentation*” at the 11th IEEE International Conference on Sustainable Technology and Engineering (i-COSTE 2025).
- Secured a **Top 5 position among 50 multidisciplinary posters** at the AUST Research Symposium 2023: An Intra-AUST Research Exhibition for undergraduate thesis research on “*Generative Adversarial Networks for Potato Disease Classification and Instance Segmentation*”.
- Recognized by **DEXIAN (Bangladesh) Limited** for **Excellence in Strategic Leadership, Agile Project Delivery, and Results-Driven Execution** in H1 2025.

Tutoring Experience

Private Academic Tutor

2020 – 2023

Mathematics, Physics & Chemistry, Grades 11–12

Dhaka, Bangladesh

- Provided one-on-one instruction in conceptual understanding, problem-solving, and examination preparation.
- Created customized lesson plans and practice materials aligned with individual learning needs, supported by regular assessment and constructive feedback.